Centenary Celebrated Sharnbasveshwar Vidya Vardhaka Sangha's



ಶರಣಬಸವ Sharnbasva ವಿಶ್ವವಿದ್ಯಾಲಯ University

ಕಲಬುರಗಿ-585 103 ಕರ್ನಾಟಕ-ಭಾರತ ಸ್ಥಾಪನೆ : 2017

Kalaburagi-585 103 Karnataka-India Estd.: 2017



Faculty of: Engineering & Technology	Department: AI&DS Program: B.Tech	AY:2026-27

INTERNAL ASSESSMENT (I.A.) TEST: I

Course Title: Fuzzy Sets, Graph Theory & Complex Variables	Course Code: 25MATS301
Date: 03/11/2026	Time: 10.00 AM To 11.30AM
Semester and Section: Third	Max. Marks:30

Note: Answer any three full questions selecting at least one question from each module

Q. No.		QUESTIONS	Marks	CO's	RBT Levels
		MODULE-I			
1	a)	Compute $y(0.1)$ *and* $y(0.2)$ by solving $\frac{dy}{dx} = \frac{y-x}{y+x}$, $y(0) = 1$ by picard's method.	6	CO1	L3
	b)	From Taylors series method, find $y(0.1)$considering up to fourth degree term if $y(x)$ satisfies the equation $\frac{dy}{dx} = x - y^2$, $y(0) = 1$	6	CO1	L3
		OR			
2	a)	Use fourth order Runge-Kutta method to compute $y(1.1)$ given that $\frac{dy}{dx} = xy^{1/3}\ y(1) = 1.$	6	CO1	L3
	b)	Apply Adam's Bashforth method to compute y *at* $x = 0.5$ correct to the three decimal places given that $\frac{dy}{dx} = \frac{x+y}{2}, y_{(0)} = 2, y_{(0.5)} = 2.636, y_{(1)} = 3.595, y_{(1.5)} = 4.968.$	6	CO1	L3
		MODULE-II			
3	a)	Let A={(x_1, 0.4), (x_2,0.3)}, B={(x_1, 0.2), (x_2 , 0.6)} verify the De-morgan's law i) $\overline{A \cup B} = \bar{A} \cap \bar{B}$ ii) $\overline{A \cap B} = \bar{A} \cup \bar{B}$	6	CO2	L3
	b)	If $A(x) = \frac{x}{x+1}, B(x) = 2^{-x}$ is the degree of membership function in fuzzy set A.Find the membership function of each of the following *i)* A^c, B^c ,*ii)* $AUB, iii)\ A \cap B, iv)(AUB)^c$.	6	CO2	L3
		OR			
4	a)	Let A, B, C are fuzzy sets defined on universal set $X = \{a, b, c, d, e\}$ as follows $A = \left\{\frac{0.2}{a} + \frac{0.5}{b} + \frac{1}{c} + \frac{0.7}{d} + \frac{0.8}{e}\right\}, B = \left\{\frac{0.3}{a} + \frac{0.6}{b} + \frac{0.9}{c} + \frac{0.4}{d} + \frac{0.2}{e}\right\}$ and $C = \left\{\frac{1}{a} + \frac{0.3}{b} + \frac{0.6}{c} + \frac{0.06}{d} + \frac{0.7}{e}\right\}$. calculate: i) (AUB) ii) (BUC) iii) (AUC) iv) (B∩ C) v) check A∪(B∩ C) =(A∪B) ∩(A∪C)	6	CO2	L3
	b)	Let A be the Fuzzy Set defined on $X = \{-1,0,1,2\}$ by membership function $A(x) = 2^{-x}$ and $f(x) = x^2$ use Extension Principle. Find $f(A)$.	6	CO2	L3

MODULE-III					
5	a)	Evaluate $\int_0^{2+i}(\bar{z})^2\ dz$ along (a) the line $x = 2y$, (b) the real axis up to 2 and then vertically to $2 + i$	6	CO3	L3
OR					
6	a)	State and prove Cauchy's theorem.	6	CO3	L2

Marks Distribution of CO's and RBT Level wise

CO's	L1	L2	L3	L4	Total Marks	% of Marks Allotted
CO1	0	0	24	0	24	40
CO2	0	0	24	0	24	40
CO3	0	6	6	0	12	20
Total RBT Marks	0	6	54	0	60	100
% of RBT Levels	0	10	90	0		